

DOCUMENT 5: IMPACT & BENEFITS DOCUMENT

Project Name: PARIKSHAK-AI (परीक्षक-AI)
Challenge: Challenge 03 — AI-Driven Examination & On-Screen Marking Transformation
Track: Technical Track | **Team Name:** eMitra

1. Executive Summary of Impact

Higher education in Madhya Pradesh encompasses **27.7 Lakh enrolled students** across **2,610+ affiliated colleges** and **~94 universities** (AISHE Data). The operational scale demands the evaluation of over **3.04 Crore handwritten answer booklets annually**.

PARIKSHAK-AI delivers systemic, measurable impact across four critical dimensions:

- 1. Educational & Student Welfare:** Compressing result turnarounds from 3–14 months down to 15–20 days, preventing students from missing corporate recruitment onboarding and competitive exam deadlines.
- 2. Citizen-Centric Governance & Institutional Trust:** Delivering responsive, transparent public service delivery under MP's Lok Seva Guarantee standards; preventing result-processing errors such as those behind BU Bhopal's ~30% suspected result discrepancies, through validation at the point of entry.
- 3. Fiscal & Economic Balancing:** Replacing a predatory ₹35+ Crore revaluation burden on distressed students with an equitable evaluation model, while saving universities ₹7.0+ Crore in paper logistics, confidential transport, and legal fees.
- 4. Equity & Linguistic Inclusion:** Giving Hindi-medium examinees (an estimated 60%+ of MP state university enrollments in general streams) equal evaluative fidelity through native multimodal reasoning.

2. Quantified Operational & Educational Metrics

Metric	Current State in MP Universities	Projected State with PARIKSHAK-AI	Operational Improvement
Result Declaration Window	90 to 420 days (3 to 14 months)	20 to 30 days post-examination	75% to 85% Turnaround Reduction
Unassessed Questions / Pages	12% to 18% of evaluated booklets	< 0.1% (Automated Page Traversal Guard)	99% Elimination of Omission Errors
Inter-Examiner Variance	Not measured during marking today	Measured continuously via blind seed scripts	Drift caught during marking, not after results
Student Revaluation Requests	6 Lakh to 9 Lakh applications / year	1.8 Lakh to 2.5 Lakh applications / year	65% to 70% Grievance Reduction
Examiner Evaluation Velocity	30–40 booklets / day (Glance-Checking)	50–65 booklets / day (Copilot Assisted)	50% Productivity Increase without Fatigue
RTI Response Generation Time	30 to 45 days (Manual Strong Room Search)	< 2 seconds (Automated PDF Dossier)	Instantaneous Administrative Defense
AI Unit Inference Cost	Not Applicable	~₹0.5–₹3 per booklet (estimate)	Small relative to logistics savings

Projected figures are targets to be validated in a university pilot; current-state figures are estimates from public reporting.

3. Comprehensive Fiscal Impact & University Net Balance Sheet

A major administrative concern for state universities (DAVV, BU Bhopal, Jiwaji, RDVV) is the displacement of revaluation fee revenue. MP state universities collect between ₹500 and ₹1,500 per paper for revaluation, generating an estimated **₹30 to ₹45 Crore annually**.

Below is our estimated net operating balance for a state university evaluating 15 Lakh answer scripts per semester (30 Lakh a year). It shows a **modest net surplus of about +₹0.7 Crore a year**; every line is an estimate to be validated in a pilot:

University Operational Balance Sheet (Annual Basis: 30 Lakh Scripts)

NET FISCAL RESTRUCTURING MODEL	
[DISPLACED REVENUE]	
• Decrease in Student Revaluation Fee Collections: (From 8% down to 2.5% revaluation application rate)	- ₹5.00 Crore
[OPERATIONAL & LOGISTICAL SAVINGS]	
• Physical Paper Logistics, Armed Police Escorts, Warehousing:	+ ₹3.20 Crore
• Second-Examiner Honorariums for Revaluation Committees:	+ ₹1.80 Crore
• Elimination of Third-Party Data-Entry Operators & OMR Scans:	+ ₹1.10 Crore
• Avoided MP High Court Legal Defense & Contempt Fines:	+ ₹0.90 Crore
[PARIKSHAK-AI OPERATIONAL EXPENDITURE]	
• AI Multimodal Inference (30 Lakh scripts @ up to ₹3):	- ₹0.90 Crore
• SDC Server & Cloud Storage Overhead:	- ₹0.40 Crore
NET ANNUAL FISCAL SURPLUS TO UNIVERSITY (ESTIMATE):	+ ₹0.70 Crore

The Examiner Incentive Realignment

Part of the logistics savings can go to examiners. Each ₹1 added per script costs about ₹30 Lakh a year at 30 Lakh scripts, so the surplus funds a modest raise; a larger raise needs separate funding:

- Raise evaluation honorariums in steps, starting with high-effort subjects.
- Faculty work with an AI copilot that reduces eye strain and tedious handwriting deciphering.
- Better pay and a lighter reading burden support faculty adoption.

4. Socio-Economic & Career Impact on Students

1. Protection of Campus Placements & Employment Onboarding:

Major corporate recruiters (TCS, Infosys, Cognizant, L&T) mandate that hired graduates produce clear, backlog-free final marks cards by July/August. Evaluation delays of 6 to 14 months routinely cause corporate offer revocations. Declaring results within 25 days safeguards thousands of engineering and commerce jobs for MP youth.

2. Eligibility for Civil & Public Service Examinations:

Aspirants appearing for UPSC CSE, MPPSC State Services, SSC CGL, and Banking (IBPS PO) require valid provisional degree certificates by strict cut-off dates. Accelerating result publication prevents candidates from losing entire annual examination cycles or crossing maximum age limits.

3. Ending the "Revaluation Tax" on Poor & Rural Students:

Students from rural tehsils in Bundelkhand, Baghelkhand, and Nimar frequently travel 200–300 km to Bhopal or Indore, staying in lodges to contest arbitrary failing marks. Reducing first-tier evaluation errors saves families thousands of rupees in avoidable travel, RTI fees, and revaluation charges.

5. Alignment with National & State Policy Frameworks

A. National Education Policy (NEP) 2020 & UGC Guidelines

- **Sections 4.34–4.37 (Transforming Assessment):** NEP 2020 mandates a shift from summative rote-testing toward competency-based evaluation. PARIKSHAK-AI's **Student Diagnostic Summaries** provide granular feedback on specific cognitive gaps rather than uninformative single-digit numbers.
- **UGC Salunkhe Committee Guidelines on Evaluation Reforms:** Directly satisfies the mandate for Outcome-Based Education (OBE), automatically mapping question marks to Course Outcomes (COs), Program Outcomes (POs), and Bloom's Taxonomy cognitive levels.
- **Section 23.2 & 23.8 (Technology Use & Artificial Intelligence):** Operationalizes the National Educational Technology Forum (NETF) directives for integrating ethical AI into high-stakes assessment workflows.

B. NAAC Accreditation Enhancement (Criterion 2: Evaluation Process & Reforms)

- **Metric 2.5.1 (Result Turnaround):** Maximizes scoring by compressing result declaration from 90–180 days down to **15–20 days**, qualifying institutions for top-tier NAAC quantitative points.
- **Metric 2.5.2 (Grievance Redressal):** Transparent Section 63 BSA dossiers cut student revaluation grievances by 65–70%.
- **Metric 2.5.3 (Examination Management Automation):** Fully integrates On-Screen Evaluation into the university's digital EMS.

C. PM-USHA (Pradhan Mantri Uchchatar Shiksha Abhiyan) Fund Unlocking

- Aligns with the **₹12,926 Crore PM-USHA national outlay**, under which state universities can seek infrastructure and IT-upgrade grants that can fund examination-branch automation.

D. MP State AI Mission (Announced March 2026)

- Aligns with **Phase 1 (2026–27) of Chief Minister Dr. Mohan Yadav's State AI Mission**, establishing scalable public-sector AI infrastructure.
- Student feedback summaries can be translated with **Digital India BHASHINI** services (planned).

E. Digital Personal Data Protection (DPDP) Act 2023 & MeghRaj Standards

- **Data Sovereignty:** In production, answer scripts, audit trails and signatures are hosted in the **MP State Data Centre (SDC), Bhopal** or a MeitY-empanelled Indian cloud, and AI inference runs on a model hosted there or in an India-region cloud. (Our proof of concept calls Google's Gemini API.)
- **Data Minimization:** Role-based access ensures examiners see only masked fictitious codes, never candidate identities.